Solution Requirements Specification

Date	15 March 2026
Team ID	Rahul-DS633
Project Name	CrimeVision AI - Multimodal Crime Scene Reconstruction
Maximum Marks	5 Marks

1. Functional Requirements (FR)

Req ID	Requirement Title	Detailed Specification & Acceptance Criteria
FR-01	Multimodal Evidence Ingestion	The system shall accept JPG/PNG crime scene images, WAV/MP3 witness and 911 audio files, and MP4/AVI surveillance videos with client-side drag-and-drop validation.
FR-02	Real-Time Object & Weapon Detection	The system shall execute RT-DETR object detection to identify humans, weapons, furniture, and evidence with confidence >= 0.70 and normalized bounding coordinates.
FR-03	3D Interactive Scene Reconstruction	The system shall compute [X, Y, Z] spatial depth coordinates and render an interactive Three.js 3D mesh with OrbitControls, directional lighting, and evidence tags.
FR-04	Speech Transcription & Multilingual Translation	The system shall transcribe acoustic witness statements using Whisper and automatically output translations in English, Hindi, Telugu, and French.
FR-05	Second-by-Second Video Timeline Alignment	The system shall extract frames from CCTV footage at 1.0 Hz and synthesize a synchronized event timeline with suspicious action classification.
FR-06	AI Forensic Conversational Assistant	The system shall provide a natural language chatbot powered by Groq LLaMA-3.3 capable of answering specific temporal and evidence queries based on session data.
FR-07	Court-Admissible PDF Generation	The system shall generate professional, cryptographic SHA-256 validated PDF forensic reports containing evidence catalogs, timelines, and legal certification.

2. Non-Functional Requirements (NFR)

Category	Target Specification	Implementation & Verification
Performance & Latency	< 3.0s for 2D image analysis, < 6.0s for audio transcription, < 10.0s for 15s video timeline.	Verified using local model caching and PyTorch CPU/GPU inference optimization.
Data Durability	Zero session data loss upon server restart or browser refresh.	Dual-tier session store persisting state as JSON records in backend/data/sessions/.
Forensic Security	Non-destructive read-only evidence handling and secure filename sanitization.	Werkzeug secure_filename and SHA-256 integrity validation.
Cross-Platform Usability	Zero-client installation; runs on standard Chromium, Firefox, and Safari browsers.	Single Page Application (SPA) built with standard HTML5, CSS3, and ES6 JavaScript.